Nikolaos Louloudakis, Ph.D.

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Professional Experience

Barclays

AI Researcher & Engineer (AVP) | *October 2025 - Present*

Conducting AI-oriented R&D as a member of the distinguished engineering team on AI.

· Applying automated analysis, evaluation, and testing of AI systems.

· Enabling research collaborations with academia and enhancing AI-oriented information retrieval.

· Developing and evaluating observability solutions, sharing expertise, and guiding stakeholders.

University of Edinburgh

Research Collaborator, AI Testing | *October 2025 - Present*

Postdoctoral Researcher, AI Testing | *October 2024 - October 2025 | 2 Top-Tier Publications (ICSE, CC)*

Worked on differential and metamorphic testing of AI/ML models and compilers.

· Applied research on DNN optimizer fault detection, discovering 14 unknown bugs in ONNX Optimizer.

· Implemented a utility that allows selective model quantization and deployment.

· Co-supervised a research intern, working on autonomous driving systems.

· PC: AAAI (2024), ICSE DeepTest, AIware, TOSEM, NEUNET, EMSE (2026), ICSE AGENTS (2027)

Oracle Labs

Research Associate (Intern) | *May 2024 – August 2024*

Implemented a novel feature for Sandwood, an ML compiler for probabilistic models.

Carbon Glance

Co-Founder & Lead Technical Coordinator | *January 2022 – May 2024*

Coordinated the design and development of the company product, related to carbon regulations.

· Formed and coordinated a team of 5 talented and strong software engineers.

· Contributed in key decisions, and introduced procedures towards effective product development.

Workable

Software Engineer | *September 2018 – April 2021* | 🏆 *Workable Security Champion*

Designed, developed, and integrated services and components for an Applicant Tracking System, following the full software development lifecycle while collaborating with cross-functional teams.

· *AMP:* Implemented a compiler to automatically create hybrid, valid AMP pages from React.

· *CI/CD:* Implemented the testing pipeline for the brand design assets of the company.

· *Search:* Developed a component responsible for the integration with a third-party search service.

· I*nternationalization:* Prepared the i18n integration of resources across a variety of services.

· *Analytics:* Refined the analytics integration mechanism for all company services.

PeopleCert Education

Coding Bootcamp Instructor | February 2018 - June 2021 (Various Iterations)

Served as coding instructor to professionals of ages 20 - 50, on multiple bootcamp iterations.

Accenture

Software Engineer - Consultant | *March 2018 – August 2018*

Worked in the largest telecommunication vendor in Greece.

· Integrated services in pipelines affecting procedures related to thousands of customers.

· Developed automation utilities that increased team productivity, saving valuable team man-hours.

Hellenic Army
Software Engineer - Analyst | *June 2017 – February 2018*
Worked on 2 .NET-based Content Management Systems.

ICS-FORTH Research Foundation
Software Engineer - Graduate Research Associate | *July 2014 – April 2017*
🏆 *Graduate Scholarship* | *Best Paper Award* | *3 Publications*
Applied Research & Development on IoT environments and automation of testing.
· Implemented the database infrastructure and a number of modules for a smart educational suite.
· Contributed towards the implementation and testing of a programmable IoT environment.

StackOverflow (Volunteer)
Established Member | *September 2013 – October 2023*
SO Profile | 220+ answers | Estimated impact: 4M Users | 6K reputation points | Top 7%

Education

Ph.D., Computer Science (AI Testing), University of Edinburgh
January 2021 – November 2024 | *First-authored 5 Top-Tier Venue Publications*
Thesis: *Testing Deep Neural Networks Across Different Computational Configurations*
Supplementary Contributions: *TA in Software Testing (4 semesters, 150+ students each) | Reviewer at AAAI SRRAI track | Supervisor of 4 M.Sc. theses | Leader of a M.Sc. Beyond-Curriculum Initiative*

M.Sc., Information Systems & Human-Computer Interaction, University of Crete
March 2014 – April 2017, 9.13/10 (Excellent) | *1 Publication (+ 3 more with ICS-FORTH)*
Thesis: *AmITest: A Framework for Semi-Automated Testing of Ambient Intelligence Environments*
Also served as TA: Human – Computer Interaction (HCI), Advanced Topics in HCI

B.Sc., Computer Science & Software Engineering, University of Crete
October 2008 – February 2014, 7.49/10 (Top 3% Of Class 2008)
Thesis: *VHF Transceiver UI Components of a System for Emergency Situations*

Selected Publications

A Selective Quantization Tuner for ONNX Models
N. Louloudakis, A. Rajan | *In IEEE/ACM ICSE NIER 2026, Rio de Janeiro, Brazil*

DiTOX: Fault Detection and Localization in the ONNX Optimizer
N. Louloudakis, A. Rajan | *In ACM SIGPLAN CC 2026, Sydney, Australia (& talk at ONNX Meetup 2026)*

FetaFix: Automatic Fault Localization and Repair of Deep Learning Model Conversions
N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In EASE 2025, Istanbul, Turkey*

Exploring Robustness of Image Recognition Models on Hardware Accelerators
N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In IEEE ICST Mutation 2025, Naples, Italy*

DeltaNN: Differential Testing to Evaluate Robustness of Image Recognition Models
N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In IEEE ICSME Industry Track 2023, Bogotá, Colombia*

Fault Localization for Buggy Deep Learning Framework Conversions in Image Recognition
N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In IEEE/ACM ASE NIER Track 2023, Kirchberg, Luxembourg*

Assessing Robustness of Image Recognition Models to Changes in the Computational Environment
N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In NeurIPS ML Safety 2022, New Orleans, USA*